if A and B are independent. As an example, recall from Equation (2.5) that if A is the appearance of a 4 on a die, and B is the appearance of an even number on a die, then $P(A) = 1/6$ and $P(A|B) = 1/3$. Since $P(A|B) \neq P(A)$ we see that A and B are dependent events.

2.2 RANDOM VARIABLES

We define a random variable (RV) as a functional mapping from a set of experimental outcomes (the domain) to a set of real numbers (the range). For example, the roll of a die can be viewed as a RV if we map the appearance of one dot on the die to the output one, the appearance of two dots on the die to the output two, and so on.

Of course, *after* we throw the die, the value of the die is no longer a random variable – it becomes certain. The outcome of a particular experiment is not an RV. If we define X as an RV that represents the roll of a die, then the probability that X will be a four is equal to 1/6. If we then roll a four, the four is a realization of the RV X. If we then roll the die again and get a three, the three is another realization of the RV X. However, the RV X exists independently of any of its realizations. This distinction between an RV and its realizations is important for understanding the concept of probability. Realizations of an RV are not equal to the RV itself. When we say that the probability of $X = 4$ is equal to 1/6, that means that there is a 1 out of 6 chance that each realization of X will be equal to 4. However, the RV X will always be random and will never be equal to a specific value.

An RV can be either continuous or discrete. The throw of a die is a discrete random variable because its realizations belong to a discrete set of values. The high temperature tomorrow is a continuous random variable because its realizations belong to a continuous set of values.

The most fundamental property of an RV X is its probability distribution function (PDF) $F_X(x)$, defined as

$$F_X(x) = P(X \leq x) \tag{2.11}$$

In the above equation, $F_X(x)$ is the PDF of the RV X, and x is a nonrandom independent variable or constant. Some properties of the PDF that can be obtained from its definition are

$$\begin{aligned} F_X(x) &\in [0,1] \\ F_X(-\infty) &= 0 \\ F_X(\infty) &= 1 \\ F_X(a) &\leq F_X(b) \quad \text{if } a \leq b \\ P(a < X \leq b) &= F_X(b) - F_X(a) \end{aligned} \tag{2.12}$$

The probability density function (pdf) $f_X(x)$ is defined as the derivative of the PDF.

$$f_X(x) = \frac{dF_X(x)}{dx} \tag{2.13}$$

Some properties of the pdf that can be obtained from this definition are

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} f_X(z)\,dz \\ f_X(x) &\geq 0 \\ \int_{-\infty}^{\infty} f_X(x)\,dx &= 1 \\ P(a < x \leq b) &= \int_a^b f_X(x)\,dx \end{aligned} \tag{2.14}$$

The Q-function of an RV is defined as one minus the PDF. This is equal to the probability that the RV is greater than the argument of the function:

$$\begin{aligned} Q(x) &= 1 - F_X(x) \\ &= P(X > x) \end{aligned} \tag{2.15}$$

Just as we spoke about conditional probabilities in Equation (2.4), we can also speak about the conditional PDF and the conditional pdf. The conditional distribution and density of the RV X given the fact that event A occurred are defined as

$$\begin{aligned} F_X(x|A) &= P(X \leq x|A) \\ &= \frac{P(X \leq x, A)}{P(A)} \\ f_X(x|A) &= \frac{dF_X(x|A)}{dx} \end{aligned} \tag{2.16}$$

Bayes' Rule, discussed in Section 2.1, can be generalized to conditional densities. Suppose we have random variables X_1 and X_2. The conditional pdf of the RV X_1 given the fact that RV X_2 is equal to the realization x_2 is defined as

$$\begin{aligned} f_{X_1|X_2}(x_1|x_2) &= P[(X_1 \leq x_1)|(X_2 = x_2)] \\ &= \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_2}(x_2)} \end{aligned} \tag{2.17}$$

Although this is not entirely intuitive, it can be derived without too much difficulty [Pap02, Pee01]. Now consider the following product of two conditional pdf's:

$$\begin{aligned} f[x_1|(x_2,x_3,x_4)]f[(x_2,x_3)|x_4] &= \frac{f(x_1,x_2,x_3,x_4)}{f(x_2,x_3,x_4)} \frac{f(x_2,x_3,x_4)}{f(x_4)} \\ &= \frac{f(x_1,x_2,x_3,x_4)}{f(x_4)} \\ &= f[(x_1,x_2,x_3)|x_4] \end{aligned} \tag{2.18}$$

Note that in the above equation we have dropped the subscripts on the $f(\cdot)$ functions for ease of notation. This is commonly done if the random variable associated with the pdf is clear from the context. This is called the Chapman–Kolmogorov equation [Pap02]. It can be extended to any number of RVs and is fundamental to the Bayesian approach to state estimation (Chapter 15).

The expected value of an RV X is defined as its average value over a large number of experiments. This can also be called the expectation, the mean, or the average of the RV. Suppose we run the experiment N times and observe a total of m different outcomes. We observe that outcome A_1 occurs n_1 times, A_2 occurs n_2 times, $\cdots$, and A_m occurs n_m times. Then the expected value of X is computed as

$$E(X) = \frac{1}{N} \sum_{i=1}^{m} A_i n_i \tag{2.19}$$

$E(X)$ is also often written as $E(x)$, $\bar{X}$, or $\bar{x}$.

At this point, we will begin to use lowercase x instead of uppercase X when the meaning is clear. We have been using uppercase X to refer to an RV, and lowercase x to refer to a realization of the RV, which is a constant or independent variable. However, it should be clear that, for example, $E(x)$ is the expected value of the RV X, and so we will interchange x and X in order to simplify notation.

As an example of the expected value of an RV, suppose that we roll a die an infinite number of times. We would expect to see each possible number (one through six) 1/6 of the time each. We can compute the expected value of the roll of the die as

$$\begin{aligned} E(X) &= \lim_{N\to\infty} \frac{1}{N} [(1)(N/6) + \cdots + (6)(N/6)] \\ &= 3.5 \end{aligned} \tag{2.20}$$

Note that the expected value of an RV is not necessarily what we would expect to see when we run a particular experiment. For example, even though the above expected value of X is 3.5, we will never see a 3.5 when we roll a die.

We can also talk about a function of an RV, just as we can talk about a function of any scalar. (We will discuss this in more detail in Section 2.3.) If a function, say $g(X)$, acts upon an RV, then the output of the function is also an RV. For example, if X is the roll of a die, then $P(X = 4) = 1/6$. If $g(X) = X^2$, then $P[g(X) = 16] = 1/6$. We can compute the expected value of any function $g(X)$ as

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx \tag{2.21}$$

where $f_X(x)$ is the pdf of X. If $g(X) = X$, then we can compute the expected value of X as

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\, dx \tag{2.22}$$

The variance of an RV is a measure of how much we expect the RV to vary from its mean. The variance is a measure of how much variability there is in an RV. In the extreme case, if the RV X always is equal to one value (for example, the die is loaded and we always get a 4 when we roll the die), then the variance of X is equal to 0. On the other extreme, if X can take on any value between $\pm\infty$ with equal probability, then the variance of X is equal to ∞. The variance of an RV is formally defined as

$$\begin{aligned} \sigma_X^2 &= E[(X - \bar{x})^2] \\ &= \int_{-\infty}^{\infty} (x - \bar{x})^2 f_X(x)\, dx \end{aligned} \tag{2.23}$$

The standard deviation of an RV is σ, which is the square root of the variance. Sometimes we denote the standard deviation as σ_X if we need to be explicit about the RV whose standard deviation we are discussing. Note that the variance can be written as

$$\begin{aligned} \sigma^2 &= E[X^2 - 2X\bar{x} + \bar{x}^2] \\ &= E(X^2) - 2\bar{x}^2 + \bar{x}^2 \\ &= E(X^2) - \bar{x}^2 \end{aligned} \tag{2.24}$$

We use the notation

$$X \sim (\bar{x}, \sigma^2) \tag{2.25}$$

to indicate that X is an RV with a mean of $\bar{x}$ and a variance of σ^2.

The skew of an RV is a measure of the asymmetry of the pdf around its mean. Skew is defined as

$$\text{skew} = E[(X - \bar{x})^3] \tag{2.26}$$

The skewness, also called the coefficient of skewness, is the skew normalized by the cube of the standard deviation:

$$\text{skewness} = \text{skew}/\sigma^3 \tag{2.27}$$

In general, the ith moment of a random variable X is the expected value of the ith power of X. The ith central moment of a random variable X is the expected value of the ith power of X minus its mean:

$$\begin{aligned} i\text{th moment of } X &= E(X^i) \\ i\text{th central moment of } X &= E[(X - \bar{x})^i] \end{aligned} \tag{2.28}$$

For example, the first moment of a random variable is equal to its mean. The first central moment of a random variable is always equal to 0. The second central moment of a random variable is equal to its variance.

An RV is called uniform if its pdf is a constant value between two limits. This indicates that the RV has an equally likely probability of obtaining any value between its limits, but a zero probability of obtaining a value outside of its limits:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b] \\ 0 & \text{otherwise} \end{cases} \tag{2.29}$$

Figure 2.2 shows the pdf of an RV that is uniformly distributed between ± 1. Note that the area of this curve is one (as is the area of all pdf's).

■ EXAMPLE 2.3

In this example we will find the mean and variance of an RV that is uniformly distributed between 1 and 3. The pdf of the RV is given as

$$f_X(x) = \begin{cases} 1/2 & x \in [1, 3] \\ 0 & \text{otherwise} \end{cases} \tag{2.30}$$

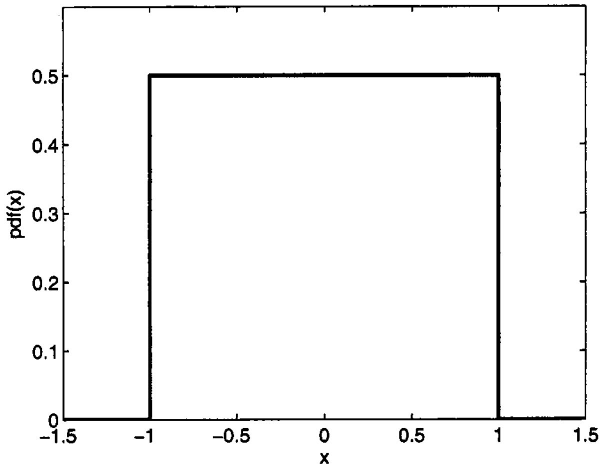

Figure 2.2 Probability density function of an RV uniformly distributed between ± 1.

The mean is computed as follows:

$$\begin{aligned}\bar{x} &= \int_{-\infty}^{\infty} x f_X(x)\,dx \\ &= \int_1^3 \frac{1}{2} x\,dx \\ &= 2\end{aligned} \tag{2.31}$$

The variance is computed as follows:

$$\begin{aligned}\sigma_X^2 &= \int_{-\infty}^{\infty} \frac{1}{2}(x-\bar{x})^2 f(x)\,dx \\ &= \int_1^3 \frac{1}{2}(x-2)^2\,dx \\ &= \frac{1}{3}\end{aligned} \tag{2.32}$$

$\nabla\nabla\nabla$

An RV is called Gaussian or normal if its pdf is given by

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(x-\bar{x})^2}{2\sigma^2}\right] \tag{2.33}$$

This is called the Laplace distribution in France, but it had many other discoverers, including Robert Adrain. Note that $\bar{x}$ and σ in the above pdf are the mean and standard deviation of the Gaussian RV. We use the notation

$$X \sim N(\bar{x}, \sigma^2) \tag{2.34}$$

to indicate that X is a Gaussian RV with a mean of $\bar{x}$ and a variance of σ^2. Figure 2.3 shows the pdf of a Gaussian RV with a mean of zero and a variance of one. If the mean changes, the pdf will shift to the left or right. If the variance increases, the pdf will spread out. If the variance decreases, the pdf will be squeezed in. The PDF of a Gaussian RV is given by

$$F_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} \exp[-(z - \bar{x})^2/2\sigma^2]\, dz \tag{2.35}$$

This integral does not have a closed-form solution, and so it must be evaluated numerically. However, its evaluation can be simplified by considering the normalized Gaussian PDF of an RV with zero mean and unity variance:

$$F_{X0}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp(-z^2/2)\, dz \tag{2.36}$$

It can be shown that any Gaussian PDF can be expressed in terms of this normalized PDF as

$$F_X(x) = F_{X0}\left(\frac{x - \bar{x}}{\sigma}\right) \tag{2.37}$$

In addition, a Gaussian PDF can be approximated as the following closed-form expression [Bor79]:

$$\begin{aligned} F_X(x) &\approx 1 - \left[\frac{1}{(1-a)x + a\sqrt{x^2 + b}}\right] \frac{\exp(-x^2/2)}{\sqrt{2\pi}} \qquad x \geq 0 \\ a &= 0.339 \\ b &= 5.510 \end{aligned} \tag{2.38}$$

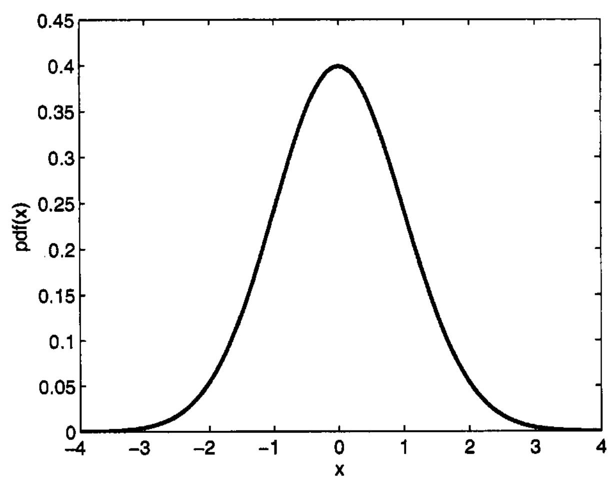

Figure 2.3 Probability density function of a Gaussian RV with a mean of zero and a variance of one.

Suppose we have a random variable X with a mean of zero and a symmetric pdf [i.e., $f_X(x) = f_X(-x)$]. This is the case, for example, for the pdf's shown in